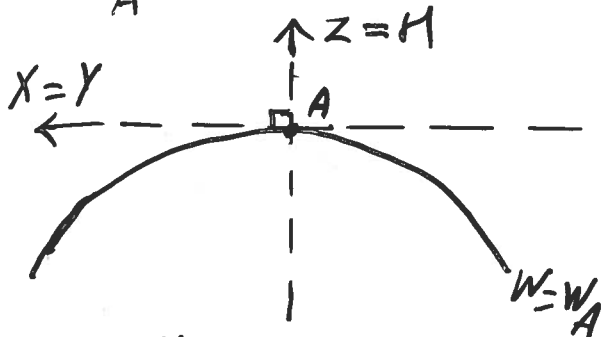


Vertical gradient of gravity

Vertical gradient of gravity, actual or normal, inside or outside the Earth, is an important quantity which will be dealt in this section. We start up with the actual gravity field. Let us consider a local right-handed cartesian system of coordinates such that the z-axis coincides with the outer normal to the equipotential surface $W=W_A$ at A.

Let the x-axis point north so that the system is equivalent to the LA



system with the direction of the y-axis reversed.

The quantity we are looking for is

$$\frac{\partial g}{\partial h} = \frac{\partial g}{\partial H} = \frac{\partial g}{\partial z} = -\frac{\partial^2 W}{\partial z^2} \quad (133)$$

Making use of Poisson's formula, the above formula can be written as

$$\frac{\partial g}{\partial h} \Big|_A = -\frac{\partial^2 W}{\partial z^2} \Big|_A = 4\pi G \sigma_A - 2\omega^2 + \left(\frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} \right) \Big|_A \quad (134)$$

where the density σ_A is evaluated according to the location of A. To interpret the last term, we use

the equation of the equipotential surface

$$W(x, y, z) = W_A$$

(135)

which can also be written explicitly as

$$z = z(x, y)$$

(136)

Following the rules for the total derivatives, the total second derivative of W with respect, for instance, to x is:

$$\frac{dW}{dx} = \frac{\partial W}{\partial x} + \frac{\partial W}{\partial z} \frac{dz}{dx} \quad z = z(x)$$

$$\frac{d^2 W}{dx^2} = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial x \partial z} \frac{dz}{dx} + \frac{\partial^2 W}{\partial z \partial x} \frac{dz}{dx} + \frac{\partial^2 W}{\partial z^2} \left(\frac{dz}{dx} \right)^2 + \frac{\partial W}{\partial z} \frac{d^2 z}{dx^2}$$

(137)

But this total derivative is equal to zero because we are dealing with an equipotential surface. Because the equipotential surface has a local extreme at A in the chosen coordinate system, even $\frac{dz}{dx}$ becomes zero, hence the above eq.

reduces to

$$y = f(x)$$

$$\text{curvature} = \frac{1}{\text{radius of curvature}} = \frac{y''}{(1+y'^2)^{3/2}}$$

$$\frac{\partial^2 W}{\partial x^2} = - \frac{\partial W}{\partial z} \frac{d^2 z}{dx^2} = g \frac{d^2 z}{dx^2}$$

(138)

From differential geometry, we know that the second derivative $\frac{d^2 z}{dx^2}$ at A is equal directly to the curvature k_{W_A} of the $z = z(x)$ profile of

(69)

the surface, since $z=z(x)$ has an extreme at A. A similar equation holds for the $z=z(y)$ profile.

By denoting the mean of curvature k_{wx} and k_{wy} by $-J = \frac{1}{2}(k_{wx} + k_{wy})$, Eq. (134) can be written as:

$$\frac{\partial g}{\partial H} = -2gJ + 4\pi G\sigma - 2\omega^2 \quad (139)$$

This equation is also called the first formula of Bruns and is valid exactly at any point in space when the proper (fractional) density is associated with points on a boundary between two media of different densities. Note that there is a discontinuity of $\frac{\partial g}{\partial H}$ at the boundary of layers of different density σ .

Equation (139) can also be applied to the normal gravity field. Limiting ourselves to the space outside the geocentric ellipsoid, where the density is zero, one obtains

$$\frac{\partial g}{\partial H} = -2gJ^N - 2\omega^2 \quad (140)$$

where J^N is the mean curvature of the corresponding

normal equipotential surface. on the reference ellipsoid, the mean curvature can be evaluated from Euler's formula simply as:

$$J_0^N = \frac{1}{2} \left(\frac{1}{M} + \frac{1}{N} \right) \quad N = \frac{a^2}{\sqrt{a^2 \cos^2 \varphi + b^2 \sin^2 \varphi}} \quad (141)$$

where M and N are the radii of curvature of the ellipsoid in the meridian and prime vertical directions. Substituting for M and N , we obtain after some development:

$$J_0^N = \frac{b}{a^2} (1 + 2f \cos^2 \varphi) \quad (142)$$

accurate to the order of e^2 . substituting back in Eq. (140) and expressing the corrective term $2\omega^2$ (much smaller than the other term) through the geodetic parameter m and γ , one finally gets the formula for the normal gravity gradient

$$\left| \frac{\partial \gamma}{\partial H} \right|_0 = - \frac{2\gamma_0}{a} (1 + m + 2f \cos^2 \varphi) \quad (143)$$

The negative sign, here as well as in Eq. (139), shows that the gradient decreases with increasing height, as would be expected. Note that this gradient is used when the normal gravity above the ellipsoid is needed.

Through a few elementary steps, Eq. (143) can be converted to

$$\begin{aligned}\frac{\partial g}{\partial H}\bigg|_0 &\doteq \frac{2g_E}{a} 1.00673(1 - 0.001415 \sin^2 \varphi) \\ &\doteq -0.308745 [\text{mGal/m}] (1 - 0.001415 \sin^2 \varphi) \quad (144)\end{aligned}$$

Taking the mean value of $\sin^2 \varphi$ to be 0.4, one gets

$$\boxed{\frac{\partial g}{\partial H}\bigg|_0 \doteq -0.3086 \text{ mGal/m}} \quad (145)$$

This is the approximate value of the gradient used in the definition of vignal's height. We leave the gradient of actual gravity in the coming section for gravimetry.

Stokes's concept

Stokes's classical approach (1849) to the determination of gravity field parameters from gravity magnitude observations is based on the solution of the external boundary value problem for the disturbing potential T . If we assume (admitting that it is so far incorrect) that there are no masses outside the geoid, then $\nabla^2 T(\vec{r}_A) = 0$ is satisfied everywhere outside the geoid. Since the value of the disturbing potential T on the ^{geoid} ellipsoid is neither known nor observable, different boundary values have to be used, i.e., a boundary value problem of a different type has to be formulated. This is where the fundamental gravimetric equation (128) is used. Using the gradient of normal gravity given by Eq. (143) and neglecting the terms of the order of flattening, one can rewrite Eq. (128) as:

$$\Delta g \doteq -\frac{2}{R}T - \frac{\partial T}{\partial H} \quad (146)$$

with R again denoting the mean radius of the Earth. This equation furnishes the boundary

values (mixed type) and can be used, in conjunction with Eq. (104), as a boundary value problem of a mixed type, involving both the function T being sought and its derivative $\frac{\partial T}{\partial H}$ both referring to the geoid. clearly, the boundary values Δg can be derived from the observed values of g and H through a few simple steps (coming in gravimetry section) and can be treated as observables. It is often referred to as the geodetic boundary value problem.

The solution to the geodetic boundary value problem is most expediently sought in the form of spherical harmonic series:

$$T(r, \theta, \lambda) = \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^{n+1} \sum_{m=0}^n \left[A_{nm}(T) \cos m\lambda + B_{nm}(T) \sin m\lambda \right] \times P_{nm}(\cos \theta) \quad (147)$$

note that $A_{nm}(T)$, $B_{nm}(T)$ are different from A_{nm} , B_{nm} in Eq. (54) but related.

Developing the boundary value Δg regarded, for the moment, to be on the sphere $r=a$ into a spherical harmonic series, one obtains:

$$\Delta g(\theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n (A_{nm}(\Delta g) \cos m\lambda + B_{nm}(\Delta g) \sin m\lambda) P_{nm}(\cos \theta) \\ = \sum_{n=0}^{\infty} \Delta g_n \quad (148)$$

where the coefficients are given by

$$\begin{Bmatrix} A_{nm}(\Delta g) \\ B_{nm}(\Delta g) \end{Bmatrix} = \frac{(n-m)!}{(n+m)!} \frac{2n+1}{2\pi} \oint_S \Delta g(\theta, \lambda) \begin{Bmatrix} \cos m\lambda \\ \sin m\lambda \end{Bmatrix} P_{nm}(\cos \theta) d\psi$$

$$A_{n0} = \frac{2n+1}{4\pi} \oint_S \Delta g(\theta, \lambda) P_n(\cos \theta) d\psi \quad (149)$$

These equations for the coefficients can easily be derived by multiplying both sides of the Eq. (148) by a certain Y_{nm}^c and Y_{nm}^s and integrating over the unit sphere. The above equations show that the coefficients A_{nm} , B_{nm} can be determined by integration of the observed values of g (Δg computed with this observation) over the Boundary. The number of coefficients A_{nm} and B_{nm} to be determined (i.e. the higher limit of n) depends on the density of observations on the Boundary. A rule of thumb shows this relation

$$n_{\max} = \frac{180^\circ}{n^\circ} \quad (150)$$

i.e. a density of $30' \times 30'$ gravity values over

the boundary gives

$$n_{\max} = \frac{180^\circ}{36^\circ = 0.5^\circ} = 360$$

This means that only the coefficients up to degree (n) and order (m) 360 is useful for the computations. !!!

Now, let us to establish the relation between the two sets of coefficients $\{A_{nm}(T), B_{nm}(T)\}$ and $\{A_{nm}(Dg), B_{nm}(Dg)\}$, so that the solution to the geodetic boundary value problem can be expressed in terms of the boundary value Dg .

To do this, Eqs. (147) and (148) are substituted into (146), which yields:

$$\begin{Bmatrix} A_{nm}(Dg) \\ B_{nm}(Dg) \end{Bmatrix} = -\frac{2}{R} \begin{Bmatrix} A_{nm}(T) \\ B_{nm}(T) \end{Bmatrix} - \frac{\partial}{\partial H} \left(\frac{a}{r} \right)^{n+1} \begin{Bmatrix} A_{nm}(T) \\ B_{nm}(T) \end{Bmatrix} \quad \begin{matrix} n=0, \dots, \infty \\ m=0, \dots, n \end{matrix} \quad (151)$$

where

$$\frac{\partial}{\partial H} \left(\frac{a}{r} \right)^{n+1} = \frac{\partial}{\partial r} \left(\frac{a}{r} \right)^{n+1} = - \left(\frac{a}{r} \right)^{n+1} \frac{n+1}{r} \quad (152)$$

and on the boundary (76) we get in particular:

$$\frac{\delta}{\delta n} \left(\frac{a}{r} \right)^{n+1} = - \frac{n+1}{R} \quad (153)$$

Hence Eq. (151) is approximately :

$$\begin{Bmatrix} A_{nm}(T) \\ B_{nm}(T) \end{Bmatrix} \approx \frac{R}{n-1} \begin{Bmatrix} A_{nm}(\Delta g) \\ B_{nm}(\Delta g) \end{Bmatrix} \quad (154)$$

clearly, for $n=1$, the relation is not defined. If a geocentric, properly oriented reference ellipsoid is considered, the the coefficients for $n=1$, i.e., $A_1(T)$, $A_{11}(T)$, $B_1(T)$, $B_{11}(T)$, are all equal to zero anyway as it has been explained in page (39).

Hence, we can write :

$$T(\theta_A, \lambda_A) = -RA_0(\Delta g) + \sum_{n=2}^{\infty} \frac{R}{n-1} \sum_{m=0}^n \left[A_{nm}(\Delta g) \cos m \lambda_A + B_{nm}(\Delta g) \sin m \lambda_A \right] P_{nm}(\cos \theta_A) \quad (155)$$

The 0th order term:

$$T_0 = -RA_0(\Delta g) = -\frac{R}{4\pi} \int_S \Delta g \, dV = -R \Delta g_0 \quad (156)$$

is a constant error in T due to an improper assessment of the total mass M^N needed in the normal potential. The quantity Δg_0 denotes the

global mean of the gravity anomalies. It is worth to note that this global mean value can be used to assess the goodness of the value of the mass of the Earth. Combining Eq. (156) and (111), one gets:

$$Dg_0 = \frac{G \delta M}{R^2} \quad (157)$$

Assuming that $\delta M = 0$ and using the notation of (106), one can finally write:

$$T(\varphi_A, \lambda_A) = \sum_{n=2}^{\infty} \frac{R}{n-1} Dg_n(\varphi_A, \lambda_A) \quad (158)$$

As the solution must not contain the first-order harmonics, these are sometimes called forbidden harmonics. This is another way of saying that the solution exists only for a properly oriented geocentric ellipsoid.

This series solution for the disturbing potential (valid for a geocentric reference ellipsoid with a mass equal to that of the Earth) can now be converted to a closed form. To do so, let us

interchange the surface integration (in the expressions for coefficients $A_{nm}(\Omega)$, $B_{nm}(\Omega)$) with the summation to obtain:

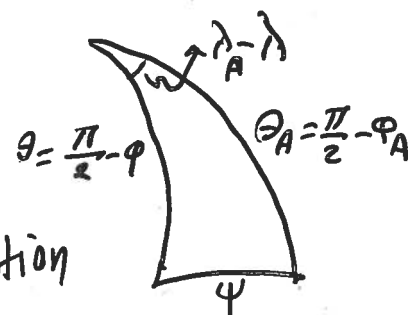
$$T(\varphi_A, \lambda_A) = \frac{R}{4\pi} \iint_S \Omega(\varphi, \lambda) \sum_{n=2}^{\infty} \frac{k(2n+1)}{n-1} \sum_{m=0}^n \frac{(n-m)!}{(n+m)!} \times \\ \times (\cos m \lambda_A \cos m \lambda + \sin m \lambda_A \sin m \lambda) P_{nm}(\sin \varphi_A) P_{nm}(\sin \varphi) d\Omega$$

$$k = \begin{cases} 1 & \text{for } m=0 \\ 2 & \text{for } m>0 \end{cases}$$

where the integration is carried over the unsubscripted arguments φ, λ on the S . Making use of Eq. (60) in page 36, one gets:

$$T(\varphi_A, \lambda_A) = \frac{R}{4\pi} \iint_S \Omega(\varphi, \lambda) \sum_{n=2}^{\infty} \frac{2n+1}{n-1} P_n(\cos \psi) d\Omega$$

where ψ is the angular distance between (φ_A, λ_A) and (φ, λ) . In Eq. (160), the series is a function of ψ only. A closed form of the series can be written



as:

$$\sum_{n=2}^{\infty} \frac{2n+1}{n-1} P_n(\cos \psi) = \overset{\text{Stokes's function}}{S(\psi)} = 1 + \frac{1}{\sin^2 \frac{\psi}{2}} - 6 \sin \frac{\psi}{2} - 5 \cos \psi - \\ - 3 \cos \psi \ln \left(\sin \frac{\psi}{2} + \sin^2 \frac{\psi}{2} \right)$$

This function is known as Stokes's function, and

its shape is as:

substituting this function

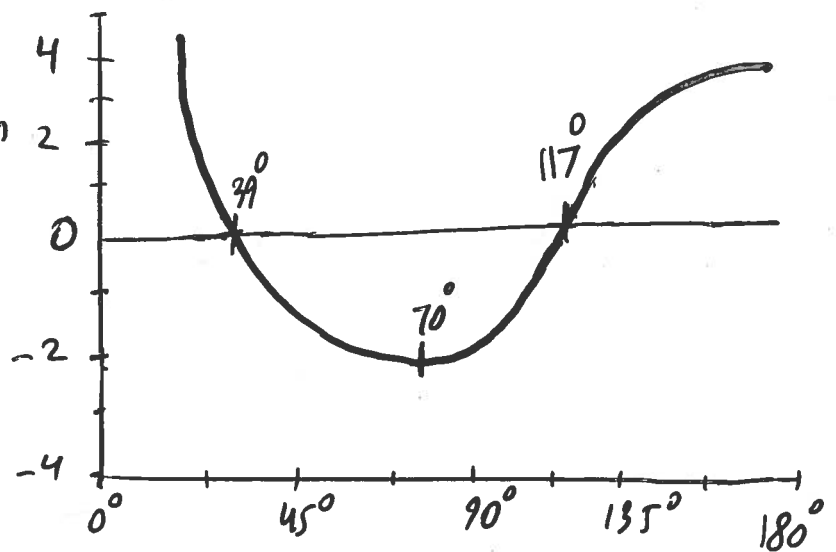
back into Eq. (160),

we finally get the

closed solution to the

boundary value problem

in the following form:



$$T(\varphi_A, \lambda_A) = \frac{R}{4\pi} \oint_S \sigma(\varphi, \lambda) S(\psi[\varphi_A, \lambda_A, \varphi, \lambda]) d\Omega \quad (162)$$

called Stokes's integral. Stokes's integral is an homogeneous and isotropic integration kernel. It is also called Green's type of solution to boundary value problem of physical geodesy.

The boundary value that σ represents must be regarded as being formulated on the geoid. Also Stokes's integral requires that there must be no masses outside the geoid. The accuracy of Stokes's integral is of the order of f (or e^2), i.e., 0.3%, due to the approximations employed. Therefore, the boundary value problem as

Formulated here is improperly posed and can be solved only approximately.

If we assume that the problem of redundant masses is eliminated, e.g., by some appropriate correction to Δg , then the disturbing potential given by Eq. (162) can be transformed to the geoidal height simply through Bruns's formula:

$$N(\varphi_A, \lambda_A) = \frac{R}{4\pi\gamma_0} \iint_S \Delta g(\varphi, \lambda) s(\psi) d\sigma \quad (163)$$

This equation is known as Stokes's formula, and it is probably the most important formula in physical geodesy. It is again a relation between the geometry and physics of the Earth. It can be shown that if the geocentric reference ellipsoid has a mass different from that of the Earth by δM , the geoidal heights become:

$$\begin{aligned} N+N_0 &= -\frac{GM}{R\gamma_0} + \frac{R}{4\pi\gamma_0} \iint_S \Delta g s(\psi) d\sigma \\ &= -\frac{R \Delta g_0}{\gamma_0} + \frac{R}{4\pi\gamma_0} \iint_S \Delta g s(\psi) d\sigma \end{aligned} \quad (164)$$

(81)

note that each milligal by which the global mean of the gravity anomalies departs from zero causes a constant error of about 6.4 m in geoidal height.

The geoidal deflection of the vertical can be obtained from the geoidal height by substitution for N in Eq. (131) of page (67) to get ?

$$\zeta(\varphi_A, \lambda_A) = - \frac{1}{4\pi\gamma_0} \iint_S \Delta g \frac{\partial S(\psi)}{\partial \varphi} \Big|_A d\psi$$

$$\eta(\varphi_A, \lambda_A) = - \frac{1}{4\pi\gamma_0 \cos \varphi_A} \iint_S \Delta g \frac{\partial S(\psi)}{\partial \lambda} \Big|_A d\psi$$

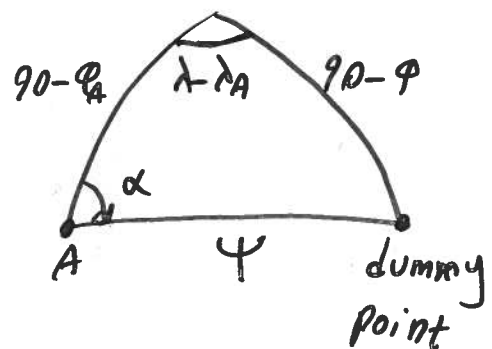
(165)

we can write ?

$$\frac{\partial S(\psi)}{\partial \varphi} = \frac{dS(\psi)}{d\psi} \frac{\partial \psi}{\partial \varphi}$$

$$\frac{\partial S(\psi)}{\partial \lambda} = \frac{dS(\psi)}{d\psi} \frac{\partial \psi}{\partial \lambda}$$

(166)



Also writing Eq. (99) in the form:

$$\cos \psi = \sin \varphi_A \sin \varphi + \cos \varphi_A \cos \varphi \cos (\lambda - \lambda_A)$$

(167)

and differentiating with respect to φ_A, λ_A , one obtains:

$$-\sin \psi \frac{\partial \psi}{\partial \varphi_A} = \cos \varphi_A \sin \varphi - \sin \varphi_A \cos \varphi \cos (\lambda - \lambda_A)$$

$$-\sin \psi \frac{\partial \psi}{\partial \lambda_A} = \cos \varphi_A \cos \varphi \sin (\lambda - \lambda_A)$$

(168)

(82)

we now introduce the azimuth α , as shown in figure of the previous page. From the spherical triangle using well known formulas of spherical trigonometry we get:

$$\sin \psi \cos \alpha = \cos \varphi_A \sin \varphi - \sin \varphi_A \cos \varphi \cos (\lambda - \lambda_A)$$

$$\sin \psi \sin \alpha = \cos \varphi \sin (\lambda - \lambda_A) \quad (169)$$

Inserting these into the preceding equations we find the simple expressions:

$$\frac{\partial \psi}{\partial \varphi_A} = -\cos \alpha \quad ; \quad \frac{\partial \psi}{\partial \lambda_A} = -\cos \varphi_A \sin \alpha \quad (170)$$

so that

$$\frac{\partial S(\psi)}{\partial \varphi_A} = -\frac{dS(\psi)}{d\psi} \cos \alpha \quad ; \quad \frac{\partial S(\psi)}{\partial \lambda} = -\frac{dS(\psi)}{d\psi} \cos \varphi_A \sin \alpha \quad (171)$$

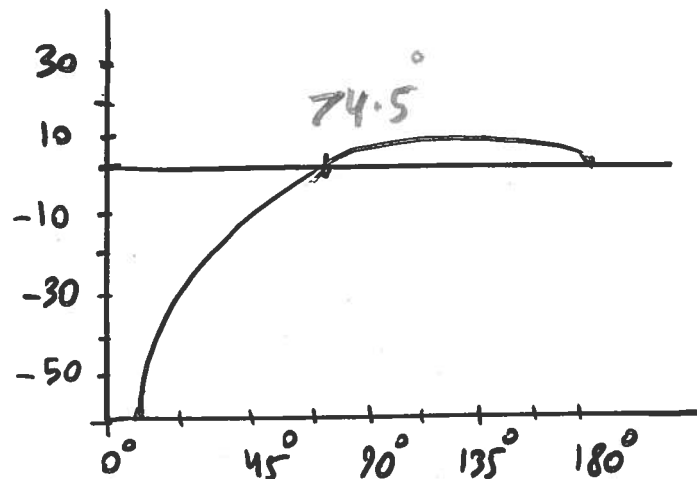
these are substituted into equation (165), one finally obtains:

$$\left\{ \begin{array}{l} \mathcal{F}(\varphi_A, \lambda_A) \\ \mathcal{L}(\varphi_A, \lambda_A) \end{array} \right\} = \frac{1}{4\pi\gamma_0} \iint_S \gamma g(\varphi, \lambda) \left\{ \begin{array}{l} \cos \alpha \\ \sin \alpha \end{array} \right\} \frac{dS(\psi)}{d\psi} d\psi \quad (172)$$

These are the formulas of Vening Meinesz. The derivative of Stokes's function is an anisotropic

kernel, known as the Vening Meinesz function, and it can be written in a closed form as:

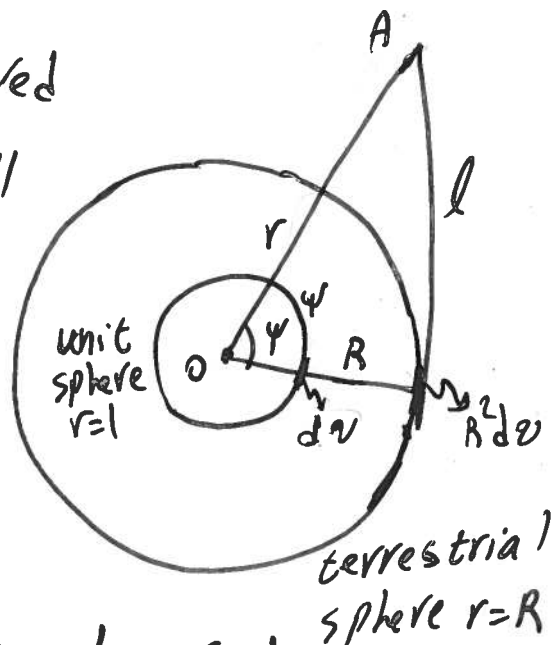
$$\frac{dS(\psi)}{d\psi} = -\frac{\cos\psi}{2\sin^2\frac{1}{2}\psi} + 8\sin\psi - 6\cos\frac{\psi}{2} - 3\frac{1-\sin\frac{\psi}{2}}{\sin\psi} + 3\sin\psi \ln\left(\sin\frac{\psi}{2} + \sin^2\frac{\psi}{2}\right) \quad (173)$$



Practical Evaluation of the integral formulas

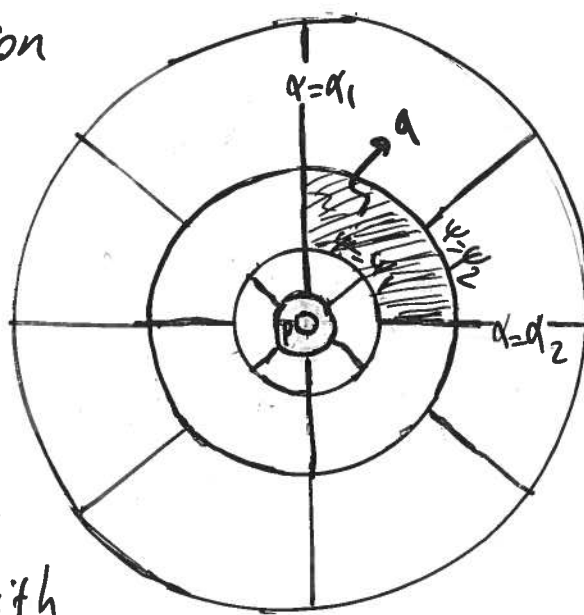
Integral formulas such as Stokes's and Vening Meinesz' formula must be evaluated over whole Earth. This means that to compute e.g. N at a point A we have to integrate Dg over the whole surface S and this reduces the importance of Stokes's integral in its current form. However, the evaluation of these integrals can be done approximately e.g. by summations.

once the values of the convolved function (Dy) are known for all the surface S , the integration can be thought of as being carried out on the surface of a unit sphere.



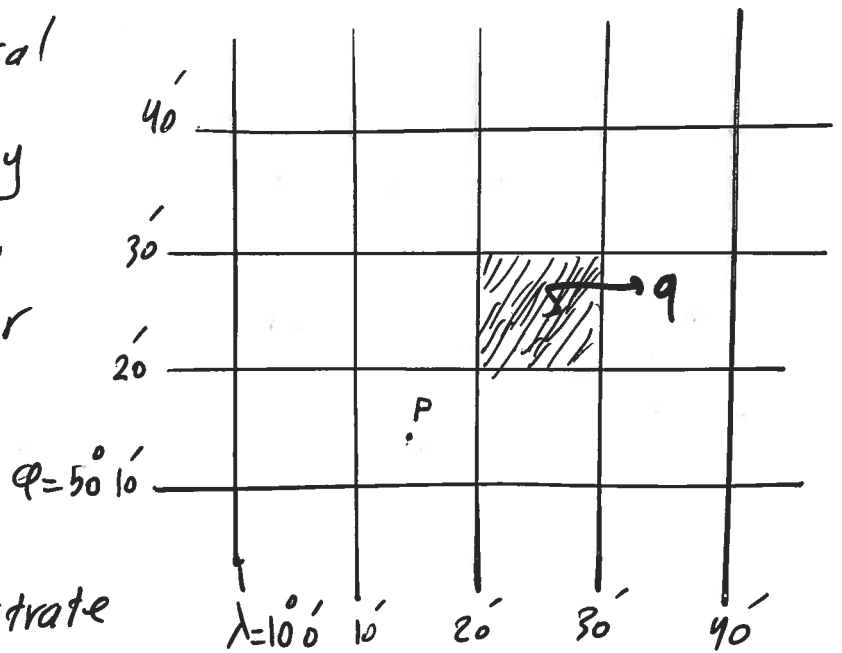
The surface elements dv can be replaced by small but finite compartments q , which are obtained by suitably subdividing the surface of the Earth. Two methods of subdivision can be used:

1- Templates: The subdivision is effected by concentric circles and their radii: the computation point p coincides with the center of the template. The natural coordinates for this template are polar coordinates ψ, α with origin at p .



2- Grid lines: The subdivision is effected by the grid lines of some fixed coordinate system, in

particular of geographical coordinates φ, λ . They form rectangular blocks, for example, of $10' \times 10'$ or $30' \times 30'$ and $1^\circ \times 1^\circ$.



As an example to illustrate the principle of numerical integration let us start with Stokes's formula:

$$N = \frac{R}{4\pi\epsilon_0} \iint_{\sigma} \Delta g S(\psi) d\sigma$$

(174)

for the template method:

$$\iint_{\sigma} = \int_{\alpha=0}^{2\pi} \int_{\psi=0}^{\pi}$$

(175)

resulting in

$$N = \frac{R}{4\pi\epsilon_0} \int_{\alpha=0}^{2\pi} \int_{\psi=0}^{\pi} \Delta g(\psi, \alpha) S(\psi) \sin \psi d\psi d\alpha$$

(176)

and for the grid method:

$$\iint_{\sigma} d\sigma = \int_{\lambda=0}^{2\pi} \int_{\varphi=-90^\circ}^{90^\circ} \cos \varphi d\varphi d\lambda$$

(177)

resulting in

$$N(\varphi, \lambda) = \frac{R}{4\pi\epsilon_0} \int_{\lambda'=0}^{2\pi} \int_{\varphi'=-90^\circ}^{+90^\circ} \Delta g(\varphi', \lambda') S(\psi) \cos \varphi' d\varphi' d\lambda'$$

(178)

(86)

In both methods for each compartment q_k the gravity anomalies are replaced by their average value $\overline{\Delta g_k}$ in this compartment. Hence the above equations can be written as:

$$N = \frac{R}{4\pi\gamma_0} \sum_k \iint_{q_k} \overline{\Delta g_k} S(\psi) d\sigma = \frac{R}{4\pi\gamma_0} \sum_k \overline{\Delta g_k} \underbrace{\iint_{q_k} S(\psi) d\sigma}_{(179)}$$

or

$$N = \sum_k c_k \overline{\Delta g_k} \quad (180)$$

where the coefficients

$$c_k = \frac{R}{4\pi\gamma_0} \iint_{q_k} S(\psi) d\sigma \quad (181)$$

are obtained by integration over the compartment q_k , and do not depend on Δg . If the Stokes's function $S(\psi)$ is reasonably constant over the compartment q_k , it may be replaced by its value $S(\psi_k)$ at the center of q_k . Then

$$c_k = \frac{R}{4\pi\gamma_0} S(\psi_k) \iint_{q_k} d\sigma = \frac{S(\psi_k)}{4\pi\gamma_0 R} \iint_{q_k} R^2 d\sigma \quad (182)$$

where the integral $\iint_{q_k} R^2 d\sigma$ is the area A_k of the

compartment. Hence one obtains:

$$c_k = \frac{A_k S(\psi_k)}{4\pi\epsilon_0 R} \quad (183)$$

This form is simpler, but close to the computation point, it may be necessary to use the integrand coefficients (181). If the compartments are formed by lines $\Delta\varphi = \text{constant}$ and $\Delta\lambda = \text{constant}$, then the computation of these integrand coefficients for the computations near the computation point might be difficult. Therefore one can use the grids for outerzone area with the formula below

$$N(\varphi, \lambda) = \frac{R}{4\pi\epsilon_0} \sum_{\Delta\varphi_i} \sum_{\Delta\lambda_j} \overline{\Delta g}_{(ij)} S(\psi)_{(ij)} \cos\varphi_{(ij)}^{2\sin\frac{\Delta\varphi_{(ij)}}{2}} \Delta\lambda_{(ij)} \quad (184)$$

and use the template method for innerzone area with the formula below:

$$N = \sum_k c_k \overline{\Delta g}_k \quad (185)$$

with c_k as below:

$$\begin{aligned} c_k &= \frac{R}{4\pi\epsilon_0} \int_{\alpha=\alpha_1}^{\alpha_2} \int_{\psi=\psi_1}^{\psi_2} S(\psi) \sin\psi \, d\psi \, d\alpha \quad (186) \\ &= \frac{R(\alpha_2 - \alpha_1)}{4\pi\epsilon_0} \int_{\psi_1}^{\psi_2} S(\psi) \sin\psi \, d\psi \quad (88) \end{aligned}$$

considering

$$F(\psi) = \frac{1}{2} S(\psi) \sin(\psi) \quad (187)$$

one can write

$$J(\psi) = \frac{1}{2} \int_0^\psi S(\psi) \sin \psi d\psi = \int_0^\psi F(\psi) d\psi \quad (188)$$

$J(\psi)$ has been tabulated and also can be programmed.

Hence, we finally obtain:

$$C_k = \frac{R(\alpha_2 - \alpha_1)}{2\pi \gamma_0} [J(\psi_2) - J(\psi_1)] \quad (189)$$

some other suggestion is to use a planar approximation in very innermost zone when $\psi \rightarrow 0$. This means:

$$S(\psi) \doteq \frac{2}{\psi} ; \quad S = R\psi \Rightarrow S(\psi) \doteq \frac{2R}{S} \quad (190)$$

one can finally obtain:

$$N_{\text{innermost zone}} \doteq \frac{S}{\gamma_0} \Delta g_p \quad (191)$$

where Δg_p is the gravity anomaly in computation point and S is the linear distance which should be

chosen properly from computation point. Actually

there are different methods to compute the geoidal height in different zones which will not be discussed here, finally resulting

$$N(\varphi, \lambda) = N_{\text{outer zone}} + N_{\text{inner zone}} + N_{\text{innermost zone}} \quad (89)$$

To complete this part, we just summarize the spherical harmonic presentation of gravity field parameters. We start with T , Eq. (108) was derived in a special form. To write T in more general form, one gets:

$$T(r, \theta, \lambda) = \frac{GM}{r} \sum_{n=2}^{n_{\max}} \left(\frac{a}{r}\right)^n \sum_{m=0}^n (\bar{R}_{nm} \cos m\lambda + \bar{q}_{nm} \sin m\lambda) \bar{P}_{nm}(\cos \theta)$$

where

$$\bar{R}_{nm} = \begin{cases} \bar{J}_{nm} - \bar{J}_n^{(N)} & \text{for } m=0 \\ \bar{J}_{nm} & \text{for } m \neq 0 \end{cases} \quad \bar{q}_{nm} = \bar{k}_{nm}$$

$$\bar{J}_n^{(N)} = \begin{cases} 0 & \text{for odd } n \\ (-1)^{n/2} \frac{3e^n (1 - \frac{n}{2} + \frac{5}{2} \frac{J_2}{e^2} n)}{(n+1)(n+3)\sqrt{2n+1}} & \text{for even } n \end{cases}$$

(192)

$$\text{with } \bar{J}_2^{(N)} = \frac{-J_2^{(N)}}{\sqrt{5}} \quad ; \quad J_2^{(N)}(\text{GRS80}) = 0.108263 \times 10^{-2}$$

$$e^2(\text{GRS80}) = 0.006694380023$$

It should be noted that in the formula for T , we have used here: $\begin{cases} J_{nm} = + \frac{a A_{nm}}{GM} \\ k_{nm} = + a \frac{B_{nm}}{GM} \end{cases}$. The sign changes if we use the minus case.

Also formula (192) is different from Eq. (109). It is because here we have used the $\oplus$ case and normalized coefficients.

considering the spherical presentation of T in the following form

$$T(r, \theta, \lambda) = \frac{GM}{r} \sum_{n=2}^{n_{\max}} \left(\frac{a}{r}\right)^n \sum_{m=0}^n (\bar{R}_{nm} \cos m\lambda + \bar{q}_{nm} \sin m\lambda) \bar{P}_{nm}(\cos \theta)$$

(193)

considering

$$\Delta g = \sum_{n=2}^{\infty} \frac{n-1}{R} T_n$$

(194)

we get:

$$\Delta g(r, \theta, \lambda) = \frac{GM}{rR} \sum_{n=2}^{n_{\max}} (n-1) \left(\frac{a}{r}\right)^n \sum_{m=0}^n (\overline{R_{nm}} \cos m\lambda + \overline{q_{nm}} \sin m\lambda) \overline{P_{nm}}(\cos \theta)$$

(195)

considering Bruns's formula on the geoid $r=R$

$$N(R, \theta, \lambda) = \frac{GM}{R\gamma_0} \sum_{n=2}^{n_{\max}} \left(\frac{a}{R}\right)^n \sum_{m=0}^n (\overline{R_{nm}} \cos m\lambda + \overline{q_{nm}} \sin m\lambda) \overline{P_{nm}}(\cos \theta)$$

(196)

The deflection of vertical can be computed from N by:

$$\xi = \frac{1}{R} \frac{\partial N}{\partial \varphi} ; \quad \eta = -\frac{1}{R \cos \varphi} \frac{\partial N}{\partial \lambda}$$

(197)

one gets:

$$\xi = \sum_{n=2}^{n_{\max}} \sum_{m=0}^n (\overline{R_{nm}} \cos m\lambda + \overline{q_{nm}} \sin m\lambda) \frac{\partial \overline{P_{nm}}(\cos \theta)}{\partial \varphi}$$

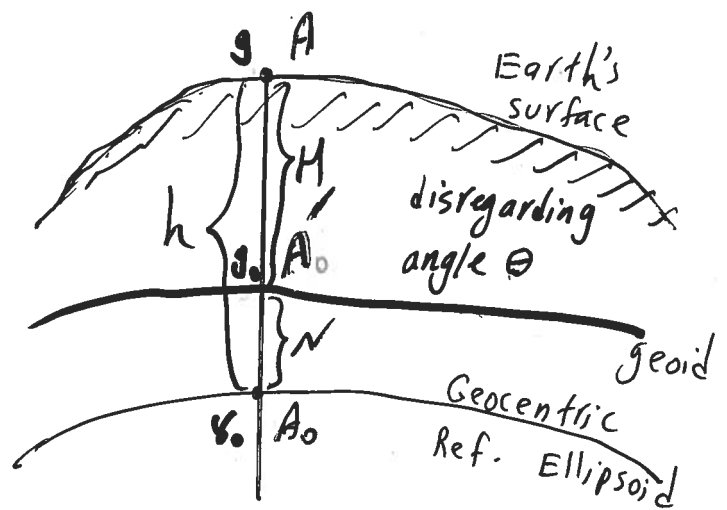
(198)

$$\eta = \sum_{n=2}^{n_{\max}} \sum_{m=0}^n (\overline{R_{nm}} \cos m\lambda + \overline{q_{nm}} \sin m\lambda) \frac{m \overline{P_{nm}}(\cos \theta)}{\cos \varphi}$$

Gravimetry

Gravity reduction:

The use of Stokes's formula for the determination of the geoidal height from gravity anomaly data requires that the gravity anomalies Δg



represent boundary values at the geoid, which implies two conditions: first, gravity g must refer to the geoid; second there must be no masses outside the geoid. Therefore, gravity reduction consists of the following steps: the topographic masses outside the geoid are completely removed or shifted below sea-level (geoid); then the gravity point is continued downward from the Earth's surface (point A) to the geoid (point A'). Depending on the way in which these topographic masses are dealt with, different gravity reduction methods are proposed. For a theoretically correct reduction

of gravity to the geoid we need the vertical gradient of gravity, $\frac{\partial g}{\partial H}$. If g is the observed value at the surface of the Earth, then the value g_0 at the geoid may be obtained as a Taylor expansion:

$$g_0 = g - \frac{\partial g}{\partial H} H + \dots \quad (199)$$

where H is the elevation of the gravity station above the geoid.

Free-air reduction:

This method of reduction assumes that there are no masses above the geoid, or that such masses have been removed beforehand, so that this reduction is indeed carried out in free-air. Neglecting all terms but the linear one in Eq. (199), one gets

$$g_0 = g + \delta g_F ; \quad \delta g_F = - \frac{\partial g}{\partial H} H \quad (200)$$

where δg_F is the free-air reduction. considering the gravity magnitude to be equal to

$$g = \frac{GM}{r^2} \quad (201)$$

Differentiation with respect to r yields:

$$\frac{\partial g}{\partial r} = \frac{\partial g}{\partial H} = -\frac{2GM}{r^3}$$

(202)

Therefore

$$\delta g_F = -\frac{\partial g}{\partial H} H = \frac{2GM}{r^3} H \xrightarrow{r=R} \delta g_F = 0.3086 H \frac{\text{mGal}}{\text{m}}$$

(203)

for mean radius values of $r=R$ and constant GM .
 H is in meters. Thereafter, we can estimate
 the free-air gravity anomaly from:

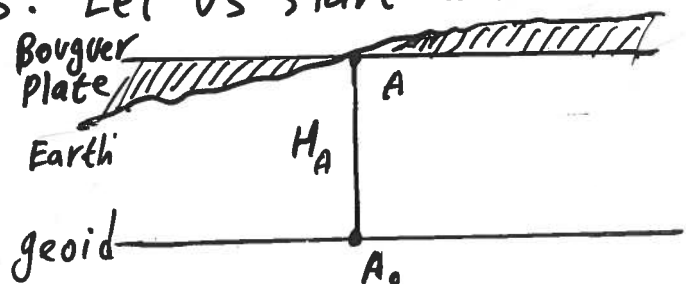
$$\Delta g_F = g_0 - \gamma_0 = g + \delta g_F - \gamma_0 = g + 0.3086 H - \gamma_0$$

(204)

The Bouguer reduction:

The object of the Bouguer reduction of gravity
 is the complete removal of the topographic masses
 , that is, the masses outside the geoid. This can

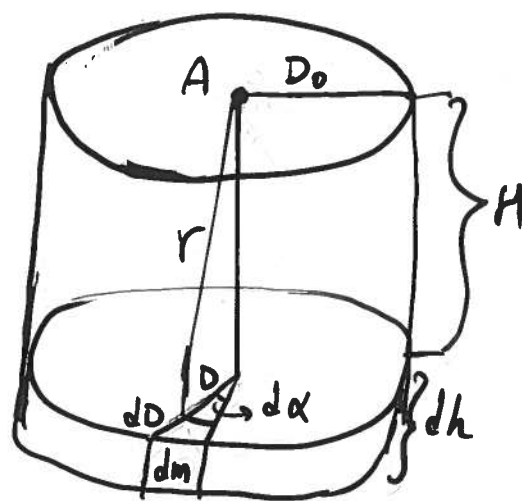
be done in two steps: 1) consider the effect of
 the Bouguer plate and 2) Implement the effect of
 remaining terrain effects. Let us start with the
 effect of Bouguer plate.



(94)

We assume that the masses between the geoid and earth's surface have a constant density σ . Now we are ready to derive an expression for the attraction of a plate of infinite extent. This step assumes a flat Earth. There are different ways to derive the attraction due to this Bouguer plate.

Referencing to this figure, the potential at A resulting from a mass element, dm , on a disk of thickness (plate) dh can be written as:



$$U = \frac{G dm}{r}$$

(205)

where $dm = dh D d\alpha dD \sigma$. Also, $r^2 = D^2 + H^2$, from which we note $r dr = D dD$. Thereafter, one can write $dm = dh r dr d\alpha \sigma$. Therefore, with this dm the potential expression for the entire disk of thickness dh and radius D_0 becomes:

$$\begin{aligned} U_{dh} &= G \sigma dh \int_0^{2\pi} \int_H^{\sqrt{H^2 + D_0^2}} d\alpha dr \\ &= G \sigma dh 2\pi \left[(H^2 + D_0^2)^{1/2} - H \right] \end{aligned}$$

(206)

(95)

The attraction of this disk is:

$$F_{dh} = -\frac{dU_{dh}}{dh} = 2\pi G \sigma dh \left(1 - \frac{h}{(h^2 + D_0^2)^{1/2}}\right) \quad (207)$$

now, the attraction of the entire disk of thickness h is:

$$F_h = 2\pi G \sigma \left[\int_0^h dh - \int_0^h \frac{h dh}{(h^2 + D_0^2)^{1/2}} \right] = 2\pi G \sigma \left[h - (h^2 + D_0^2)^{1/2} + D_0 \right] \quad (208)$$

This is the attraction of the disk for a point on the disk. However, later we need the attraction for a point outside the disk, e.g. with a height $h + \delta h$. The result would be:

$$F_{h+\delta h} = 2\pi G \sigma \left[h + (\delta h^2 + D_0^2)^{1/2} - ((h + \delta h)^2 + D_0^2)^{1/2} \right] \quad (209)$$

It is obvious that for a point on the disk Eq (209) becomes equal to Eq (208), putting $\delta h = 0$.

For a disk of infinite radius $D_0 \rightarrow \infty$, h is assumed small relative to D_0 , and Eq. (208) may be written:

$$F_h = 2\pi G \sigma h \quad (210)$$

with the standard density $\sigma = 2.67 \frac{g}{cm^3}$, one finally gets:

$$F = \sigma g_p = 0.1119 h \left| \frac{mGal}{m} \right| \quad h \text{ in meters} \quad (211)$$

σg_p is the attraction of Bouguer plate. After applying the attraction of plate with subtracting it from the observed gravity, one must continue downward the gravity station A to A_0 , which can be done by applying the free-air reduction (96)

Therefore we can write:

$$g_p = g - \delta g_p + \delta g_F = g + 0.3086H - 0.1119H$$

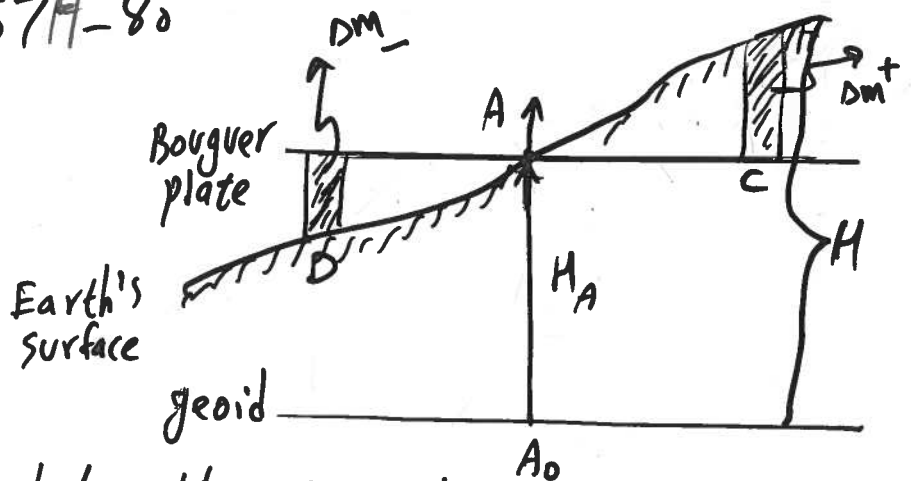
$$[g_p = 0.1967H] \frac{mGal}{m} \quad (212)$$

This is called simple Bouguer gravity. Thereafter, we can write the simple Bouguer gravity anomaly as:

$$\begin{aligned} \Delta g_p &= g_p - g_0 = g - \delta g_p + \delta g_F - g_0 \\ &= g_p + 0.1967H - g_0 \end{aligned} \quad (213)$$

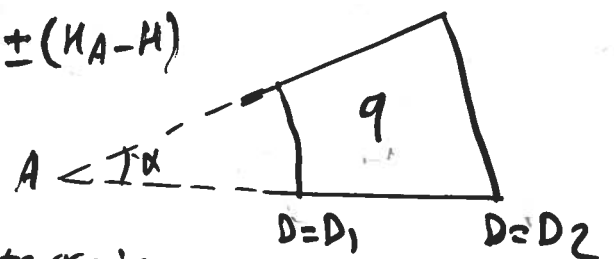
Second step:

Now, we have considered the effect of Bouguer plate, and to complete the removal of masses above the geoid we have to take into account the deviation of the actual topography from the Bouguer plate of point A. This is called terrain correction. At C part the mass surplus (redundancy) Δm_+ , which attracts upward, is removed, causing g at A to increase. At D part the mass deficiency Δm_- is made up, causing g



at A to increase again. The terrain correction is always positive. The practical determination of the terrain correction F_t is done by means of a template as shown before. The attraction of the terrain can be expressed by referring to Eq. (209). If H_A is the elevation of the point being reduced, and H is the mean elevation of each compartment in the template method, and if the width of the ring or zone is $D_2 - D_1$ and there are n compartments, and $\Delta H = \pm(H_A - H)$

$\alpha = \frac{2\pi}{n}$, then the vertical



attraction of the compartment terrain on

the point of computation is

$$F_t|_q = \delta g_t|_q = -\frac{2\pi G\sigma}{n} \left[(D_1 + \sqrt{D_2^2 + \Delta H^2} - D_2 - \sqrt{D_1^2 + \Delta H^2}) \right] \quad (214)$$

Adding the effects of the individual compartments one finally obtains:

$$F_t = \delta g_t = \sum F_t|_q = \sum \delta g_t|_q \quad (215)$$

Note that the terms in square brackets tend to cancel out as D increases. Thus we do not have to carry the

integration beyond a few tens of kilometers.

This is only true by the presentation of terrain effects by this type of formula, where curvature of the Earth is not considered. In addition, if dh is equal to zero everywhere, the effect of δg_t is also equal to zero, as expected. In reality values reaching over a hundred milligal are experienced in mountains. By adding the terrain correction δg_t to Eqs. (212) and (213) we obtain the refined Bouguer gravity and gravity anomaly

$$g_B = g + \delta g_F - \delta g_P + \delta g_t \quad (216)$$

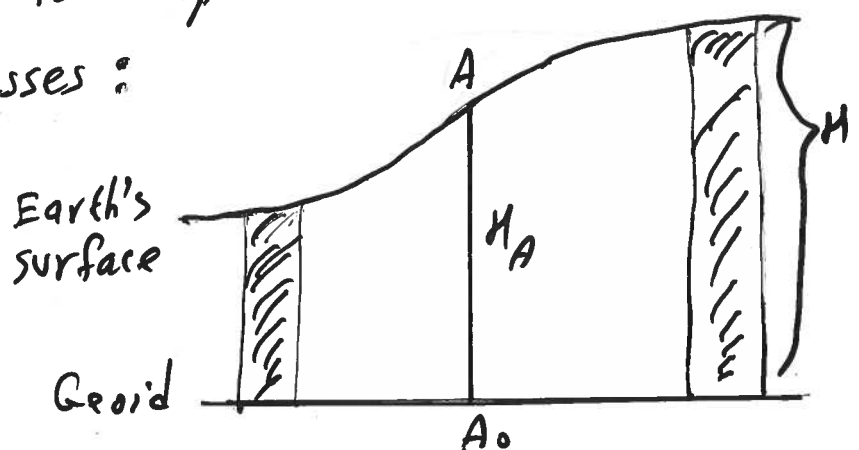
corresponding gravity anomaly (refined Bouguer) may be written as;

$$\Delta g_B = g + \delta g_F - \delta g_P + \delta g_t - g_0 \quad (217)$$

It is also possible to compute the total effect of the topographic masses:

$$\delta g_T = \delta g_P - \delta g_t \quad (218)$$

in one step by using columns with base at sea level, again



subdividing the terrain by means of a e.g. template. note the difference between δg_T , the attraction of the topographic masses, and the terrain correction δg_t . Then

$$\delta g_T = \sum \delta g_{T|q} \quad (219)$$

where

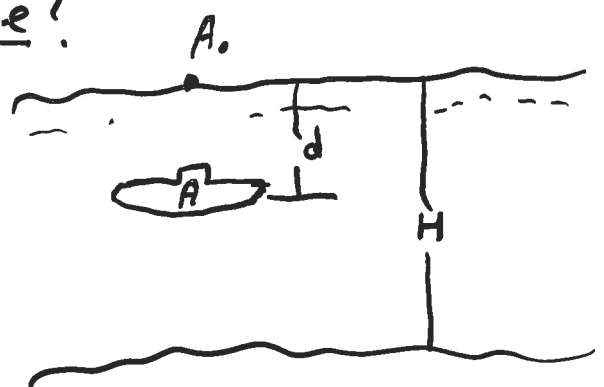
$$\delta g_{T|q} = -\frac{2\pi G \rho}{n} \left[\sqrt{D_2^2 + H_A^2} - \sqrt{D_1^2 + H_A^2} - \sqrt{D_1^2 + \Delta H^2} + \sqrt{D_2^2 + \Delta H^2} \right] \quad (220); \Delta H = H_A - H$$

therefore, instead of Eq. (217) we can write:

$$\Delta g_B = g + \delta g_F - \delta g_T \quad (221)$$

now what happens with gravity observation made for example aboard a submarine?

Actually gravity observations on a submarine have quite similar reductions to those made on land. we just summarize the result of reductions:



$$\delta g_F^{\text{sea}} = 0.3086 d \frac{\text{mGal}}{\text{m}} \quad \text{for } d \text{ in meters} \quad (222)$$

The correction of free-air must be subtracted from gravity observation in A, i.e.:

There after the free-air gravity anomaly for observations made at sea can be written as:

$$\Delta g_F^{\text{sea}} = g - \delta g_F^{\text{sea}} \pm \delta g_{\text{Eöt}} - \gamma_0 \quad (223)$$

where $\delta g_{\text{Eöt}}$ is a different type of correction and is considered to gravity observations made from a moving vehicle. The Eötvös correction is necessary to account for the east-west component of the velocity of the vehicle (here submarine). This correction is expressed as:

$$\delta g_{\text{Eöt}} = \pm 2\omega v \cos \varphi \quad (224)$$

where v is the east-west velocity component, φ is the geodetic latitude, and ω is the rotational velocity of the Earth. The Eötvös correction is always positive for eastward travel and negative for westward travel.

The Bouguer plate correction for the observations made at sea accounts for the thin plate of water of infinite extent which is above the submarine. This correction must be applied twice, once to remove the effect of the upward pull of the plate of water

and again to restore the effect of the downward pull of the plate of water if the gravimeter had been on the surface. The reduction is always positive :

$$\delta g_{BP}^{\text{sea}} = 2\pi G \sigma_{\text{wat}} d \quad (225)$$

where σ_{wat} is the density of sea water.

It should be mentioned here that there is a second Bouguer correction for the observations made at sea. This correction accounts for the effect of the slab of water of thickness h' (Fig. page 100) which extends to the ocean floor. This correction is based on the assumption that the ocean basins represent a mass deficiency not compensated

isostatically. It is necessary, according to Bouguer, to remove the mountains completely (which was done with Bouguer correction on land) and, similarly, to carry additional masses to the oceans are already filled with material of density 2.67 g/cm^3 . Because the oceans are already filled with material of density σ_{wat} , It is necessary to consider only

the density difference ($\sigma - \sigma_{wat}$). This correction is always positive:

$$\delta g_{BP2}^{sea} = 2\pi G (\sigma - \sigma_{wat}) H' \quad (226)$$

Finally the Bouguer gravity anomaly for the observations made on a submarine can be written

$$\Delta g_B^{sea} = g - \delta g_F^{sea} + \delta g_{BP}^{sea} + \delta g_{BP2}^{sea} \pm \delta g_{Eöt} - \gamma_0 \quad (227)$$

for the reduction of gravity observed from an aircraft the situation is the same as on land.

Free air, Bouguer, Isostatic, Topographic corrections are the same as on land. The Eötvös correction must also be included.

Isostasy

one might be inclined to assume that the topographic masses are simply superposed on an essentially homogenous crust. If this were the case, the Bouguer reduction would remove the main irregularities of the gravity field, so that the Bouguer anomalies would be very small and would fluctuate randomly around zero. However, just the opposite is true. Bouguer anomalies in mountainous areas are systematically negative and may attain large values, increasing in magnitude on the average by 100 mGals per 1000 meters of elevation. The only explanation possible is that there is some kind of mass deficiency under the mountains. This means that the topographic masses are compensated in some way.

There is the similar effect for the deflection of the vertical. The actual deflections are smaller than the visible topographic masses would suggest. one such an observation by Pratt in the Himalays

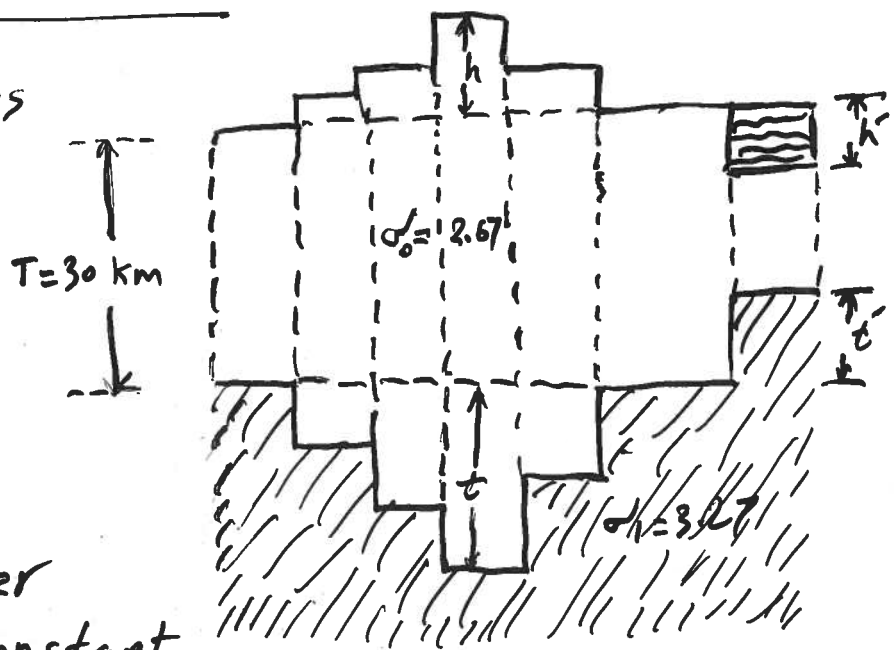
computed a value of $28''$ for the deflection of the vertical, while the measurements show $5''$. Again, some kind of compensation is needed to account for this discrepancy. Two different theories for such a compensation were developed by Pratt and by Airy. According to Pratt the mountains have risen from the underground somewhat like a fermenting dough. According to Airy, the mountains are floating on a fluid lava of higher density, so that the higher the mountain, the deeper it sinks. Here, we discuss and present Airy theory.

Airy - Heiskanen system:

This Figure shows the principle. The mountains, of constant density

$$\sigma_0 = 2.67 \frac{\text{g}}{\text{cm}^3}$$

float on a denser underlayer of constant



density

$$\sigma_1 = 3.27 \frac{\text{g}}{\text{cm}^3}$$

The higher they are, the deeper they sink. Thus, root formations exist under mountains, and antiroots under the oceans. We denote the density difference $\sigma_1 - \sigma_0 = \Delta\sigma$. With the assumed numerical values one obtains:

$$\Delta\sigma = 0.6 \frac{\text{g}}{\text{cm}^3}$$

If we denote the height of the topography by h and thickness of the corresponding root by t , then the condition of floating equilibrium is

$$t \Delta\sigma = h \sigma_0 \quad (228)$$

so that

$$t = \frac{\sigma_0}{\Delta\sigma} h = 4.45 h \quad (229)$$

For the oceans the corresponding condition is

$$t' \Delta\sigma = h' (\sigma_0 - \sigma_w) \quad (230)$$

where h' and σ_w are defined as above with

$$\sigma_w = 1.03 \frac{\text{g}}{\text{cm}^3} \text{ and } t' \text{ is the thickness of the}$$

antiroot, so that one gets

$$t' = \frac{\sigma_0 - \sigma_w}{\sigma_1 - \sigma_0} h' = 2.73 h' \quad (231)$$

for the numerical values assumed.

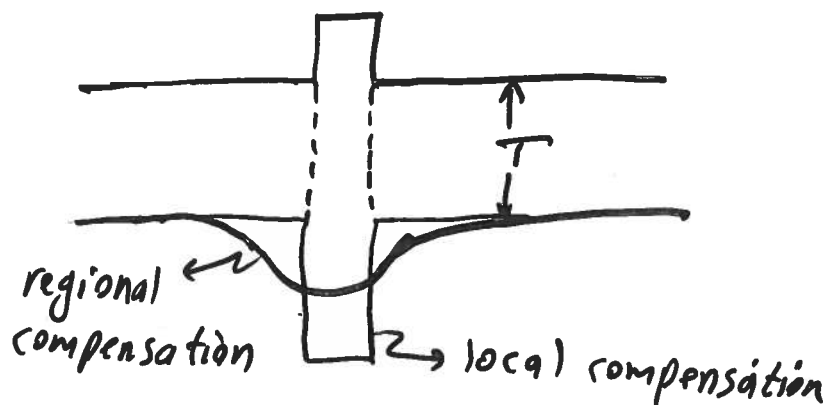
The normal thickness of the Earth's crust is denoted by T . $T = 30$ km is a value assumed. The crustal thickness under mountain is then

$$T + h + t \quad (232)$$

and under the oceans it is

$$T - h' - t' \quad (233)$$

Vening Meinesz modified the Airy floating theory introducing regional instead of local compensation.



Isostatic Reductions

The object of isostatic reduction of gravity is the regularization of the Earth's crust according to some model of isostasy. Here, the topographic masses are not completely removed

as they are in the Bouguer reduction, but are shifted into the interior of the geoid in order to make up the mass deficiencies that exist under the continents. In the Airy-Heiskanen model the topographic masses are used to fill the roots of the continents, bringing the density from $\sigma_0 = 2.67$ to $\sigma_1 = 3.27 \frac{\text{g}}{\text{cm}^3}$. In other words, the topography is removed with its compensation, and the final result is ideally a homogeneous crust of density σ_0 and constant thickness T .

Thus we have three steps:

- 1) removal of topography
- 2) removal of compensation
- 3) free-air reduction to geoid

steps 1 and 3 are known and step 2 is discussed below.

To compute the removal of compensation, we refer again to Eq. (209) with the parameters defined in Airy-Heiskanen system. one finally gets:

$$\delta g_c|_q = \frac{2\pi G \Delta \sigma}{n} \left[\sqrt{D_2^2 + (hp+T)^2} - \sqrt{D_1^2 + (hp+T)^2} - \sqrt{D_2^2 + (hp+T+t)^2} + \sqrt{D_1^2 + (hp+T+t)^2} \right]$$

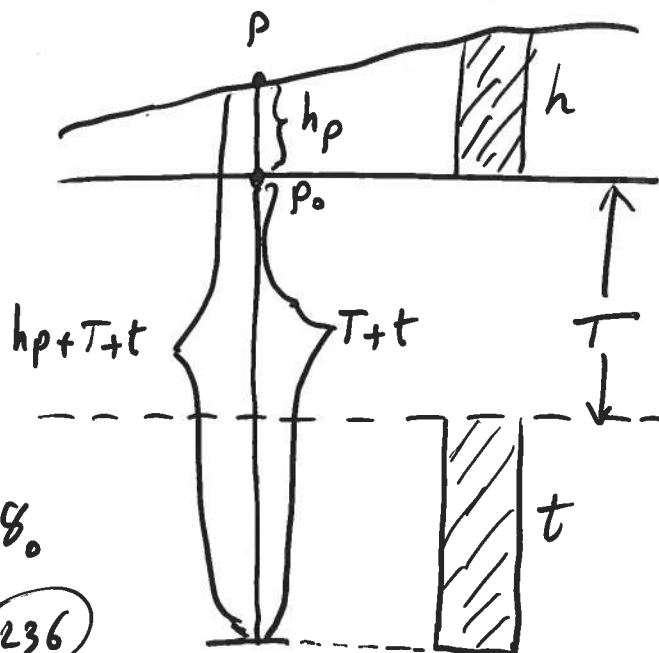
Adding the effects of the individual compartments one finally obtains:

(234)

$$\delta g_c = \sum \delta g_c|_q$$

(235)

Therefore, the Isostatic gravity anomaly can be written as (on land):



$$\Delta g_I = g - \delta g_T + \delta g_c + \delta g_F - g_0$$

(236)

where $-\delta g_c$ is the attraction of the compensation which is actually negative, so that its removal is equivalent to the term $+\delta g_c$. The δg_T is the attraction of topography, to be computed as the effect of a Bouguer plane combined with terrain correction.

(109)

For the oceanic stations the terms δg_T

and δg_F are zero, since the station is suited on the geoid, but the term δg_C is more complicated.

In the Airy-Heiskanen model the mass surplus of the antiroot, $\sigma_1 - \sigma_0$, is used to fill the oceans to the proper density σ_0 . The corresponding value is given by

$$\delta g_C^{\text{oce}} = -\delta g_{C1} + \delta g_{C2} \quad (237)$$

where both δg_{C1} and δg_{C2} have the same nature as terrain correction.

$$\delta g_{C1}|_q = \frac{2\pi G (\sigma_1 - \sigma_0)}{n} \left[\sqrt{D_2^2 + (T-t')^2} - \sqrt{D_1^2 + (T-t)^2} - \sqrt{D_2^2 + T^2} + \sqrt{D_1^2 + T^2} \right] \quad (238)$$

$$\delta g_{C1} = \sum \delta g_{C1}|_q \quad (239)$$

$$\delta g_{C2}|_q = \frac{2\pi G (\sigma_0 - \sigma_w)}{n} \left[D_2 - D_1 - \sqrt{D_2^2 + h'^2} + \sqrt{D_1^2 + h'^2} \right] \quad (240)$$

$$\delta g_{C2} = \sum \delta g_{C2}|_q \quad (241)$$

Then finally the gravity anomaly (Isostatic) for oceanic area can be written

$$\Delta g_{\text{I}}^{\text{oce}} = g + \delta g_{\text{c}}^{\text{oce}} - g_0 \quad (242)$$

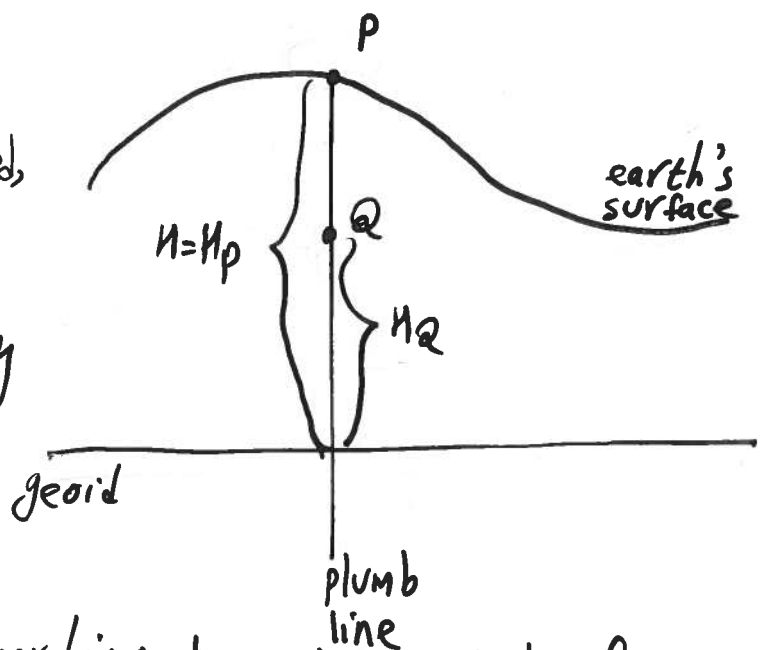
If any of the isostatic systems were rigorously true, then isostatic reduction would fulfil perfectly its object of complete regularization of the Earth's crust, which would become level and homogeneous. Then with a properly chosen reference model for g_0 , the isostatic anomalies would be zero.

The actual isostatic compensation occurring in nature cannot, of course, completely conform to such abstract models. Therefore, nonzero isostatic anomalies will be left, but they will be small, smooth, and more or less randomly positive and negative.

The gravity reduction of poicare-pray

This gravity reduction does not properly belong here because it is intended to provide actual values of gravity inside the Earth, and not to give boundary values at the geoid. It should be mentioned that this reduction is needed in height systems section. However, for sake of completeness it would be presented here.

Since g' inside the Earth cannot be measured, it must be computed from the surface gravity g_p . This is done by reducing the measured



values of gravity according to the method of poicare-pray. Let us consider that g' must be computed at point Q inside the Earth $g' = g_Q$ and the corresponding surface measurement is done in point P, then $g = g_P$. Thereafter, the computed value of g_Q can be written

$$g_Q = g_P - \int_Q^P \frac{\partial g}{\partial H} dH$$

(112)

(243)

The actual gravity gradient $\frac{\partial g}{\partial H}$ can be written as:

$$\frac{\partial g}{\partial H} = -2gJ + 4\pi G\sigma - 2\omega^2 \quad (244)$$

If the mean curvature J of the geopotential surfaces and the density σ between p and q are known. The normal free-air gradient can be obtained by:

$$\frac{\partial g}{\partial H} = -2gJ^N - 2\omega^2 \quad (245)$$

using the approximation: $gJ \approx gJ^N$

$$\frac{\partial g}{\partial H} = \frac{\partial g}{\partial H} + 4\pi G\sigma \quad (246)$$

Neglecting the variation of $\frac{\partial g}{\partial H}$ with latitude, with $\sigma = 2.67 \frac{g}{cm^3}$ and $G = 66.7 \times 10^{-9} \frac{g^{-1} cm^3}{s^2}$

$$\frac{\partial g}{\partial H} = -0.3086 + 0.2238 = -0.084 \frac{mGal}{m} \quad (247)$$

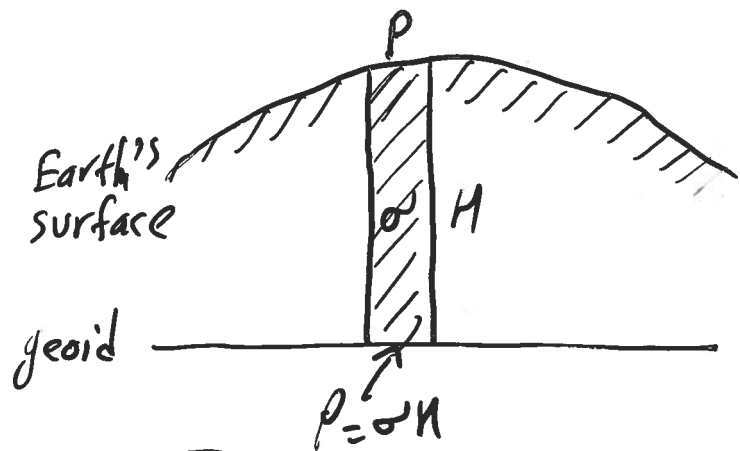
so that

$$\underline{g_Q = g_P + 0.0848 (H_P - H_Q)} \quad (248)$$

This simple formula, although some approximation made might be crude, is often applied in practice. It yields the actual gravity which would be measured inside the Earth if this were possible.

the gravity anomaly reduction using Helmert's second condensation method

Here the topography is condensed so as to form a surface layer on the geoid of density:



$$\rho = \sigma H$$

(249)

The resulting gravity on the geoid can be written as below referring to Eq. (209), letting the height of disk goes to zero:

$$g_H = g - \delta g_T + \delta g_H + \delta g_F$$

(250)

where

$$\delta g_H|_q = \frac{2\pi G \rho}{n} \left(-\frac{n p}{\sqrt{D_2^2 + n^2}} + \frac{n p}{\sqrt{D_1^2 + n^2}} \right)$$

(251)

where $n p$ is the height of the station p and H the height of the compartment. Finally

$$\delta g_H = \sum \delta g_H|_q$$

(252)

and gravity anomaly (Helmert) becomes

$$\Delta g_H = g_H - g_0 = g - \delta g_T + \delta g_H + \delta g_F - g_0$$

(253)

Indirect effect :

The removal or shifting of masses which underly the gravity reductions change the gravity potential and hence the geoid. This change of geoid is an indirect effect of the gravity reductions. Hence, the surface computed by Stokes's formula from isostatic (Helmert or free-air) gravity anomalies is not the geoid itself but a slightly different surface, called the cogeoid. To every gravity reduction there corresponds a different cogeoid.

Let the undulation of the cogeoid be N^c . Then the undulation N of the actual geoid may be obtained from:

$$N = N^c + \delta N_{Ind}$$

(254)

where δN_{Ind} is the indirect effect on geoid, which can be computed from:

$$\delta N_{Ind} = \frac{\delta W}{\gamma_0}$$

(255)

where δW is the change of potential at the geoid, which δW can be computed for every gravity reduction method in the same way as gravity attraction.

Some notes in different gravity reduction methods

First of all, it should be mentioned that all formulas for the gravity reduction are presented with Template method of practical computations. The grid line method can also be used, which has nowadays been used more often. It is left to students to investigate the use of grid line method.

To be adequate for the use in Stokes's formula, the gravity reduction should satisfy the following conditions:

- 1) No masses outside the geoid
- 2) No change of total mass
- 3) No change in locations of the center of gravity
- 4) The indirect effect should not be unduly great

The gravity reductions generally must satisfy following requirements:

- 1) The gravity reductions must yield gravity anomalies that are small and smooth, so they can be easily interpolated and extrapolated. They should

be as representative as possible of a whole neighborhood.

2) The reductions must correspond to a geophysically meaningful model, to be useful for geophysical and geological interpretations.

The Bouguer anomaly satisfies the condition that no masses must be outside the geoid. They also have a good interpolatory properties (they are large but smooth) and are geophysically significant.

But, to a considerable degree, changes the Earth's mass and the center of gravity and their indirect effect might be hundreds of meters and therefore cannot be used in Stokes's formula.

The free-air reduction and Helmert condensation reduction do not change the total mass and no masses outside geoid are in these models. They have small (a few meters) indirect effects and are simple to compute. Both change the center of gravity slightly and suited for the use in Stokes's formula for geoid determination. However, free-air anomalies are small but extremely

dependent on topography, so that their interpolation is very inaccurate.

The isostatic correction replaces the mass removed by the Bouguer reduction and replaces it in a manner consistent with the actual condition on the Earth. No masses outside the geoid and no change of total mass are the result of this reduction. They slightly change center of gravity and their indirect effect is moderate. Hence, the isostatic anomalies must be considered best suited to be used in the Stokes's formula for geoidal height determination. The isostatic anomalies correspond best to geological reality. They are small, smooth and independent of topography, so that they are ideally suited for interpolation and extrapolation and are very representative.

To illustrate the magnitude of the anomalies resulting from different reductions under different observation conditions an example is shown in next page.

Gravity anomalies in mGal

	H (m)	$\Delta g_f + \Delta g_t$	Δg_B	Δg_I (T=40 km) Airy-Heiskanen
Mountain terrain				
Las Vegas	1960	+36	-181	+8
Truckee	1805	+45	-154	-4
Grand canyon	2386	+52	-200	0
Lake placid	571	+38	-9	+16
Valley Terrain				
Knoxville	280	-6	-37	-9
El Paso	1146	+24	-103	+29
Colorado springs	1841	+2	-180	+8
salt lake city	1322	-15	-138	+26
Sea				
				T=30 km
Mediterranean	-2200	-1	+149	+8
Indian ocean	-4390	-33	+266	-30
North Atlantic	-3610	+26	+272	+36
South Atlantic	-3700	+28	+280	+26
Pacific ocean	-4930	-6	+329	+5

It is apparent that the Bouguer anomalies are negative on land and positive in oceans, and are generally much larger than either free-air or isostatic anomalies. Assuming that the isostatic correction is a valid representation of

of the truth, as geophysical and geodetic evidence shows that the Earth is about 90% isostatically compensated, this result is to be expected.

Assuming that the root of lighter material extending under mountains, then the measured value of gravity on the mountain top would be less than if the mountain were merely excess material piled on top of a crust of uniform density and thickness. Because the Bouguer correction does not consider this condition, the less-than-anticipated value of gravity is reduced uncompensated and result in a negative anomaly. The opposite reasoning may be applied to the situation in ocean areas.

Further, consider the meaning of zero gravity anomalies the world over. If all free-air anomalies were zero, all topography above the geoid would be without mass. For all Bouguer anomalies to be zero, mountains would indeed be excess piles of rubble on the geoid. The only zero anomaly

conditions with more than trivial meaning are those based on isostasy—in this case, they mean that the earth is in the equilibrium hypothesized for the reduction. positive isostatic anomalies indicate undercompensation; such would be the case on a volcanic island so small it can be partially supported by the rigidity of the crust. For example, Cyprus has isostatic anomalies of 100 mGal. A negative anomaly means overcompensation, as in the case of a mountain wearing down by erosion faster than the compensating material disappears below it. From the gravity anomalies shown in page 119, it appears that the free-air anomalies are close to those that result from isostasy with $T=40$ km. This consistency with the more adequate but rigorous isostatic reduction and its ease of application leads to this fact that the terrain corrected free-air gravity anomalies and Helmert anomalies be used instead of isostatic anomalies in Stokes's formula

Gravity data :

The previous sections showed how gravity data are converted into other parameters of the Earth's gravity field. So far, nothing has been said about how the gravity data is acquired and processed to obtain a consistent set of global anomalies appropriate for use in different formulas discussed earlier.

Theoretically, gravity anomalies must be given at every point of the Earth's surface; particularly a dense gravity net around the computation points is needed. A reasonably uniform distribution of gravity measurements outside are sufficient.

It should be noted that all gravity anomalies must be converted to the same system.

Gravity at a point on the surface of the Earth can be determined directly by either pendulums or free-fall devices. It is sufficient here to say that, in both cases, precise timing of movements is required, and the observed time interval is subsequently converted into the value of g using the appropriate equation of motion. The achievable accuracy,

dependently largely on control of the instrument environment and the number of repeated observations, is of the order of some μGal in the best case and some tenth of μGal in worst case. Both kinds of instruments give us absolute gravity measurements. more portable, handier to operate, cheaper, and thus more widely used are gravimeters, instruments capable of measuring only differences in gravity between pairs of points. There are many types of gravimeters making use of various principles. Most designs use the basic idea of a small mass suspended from a responsive element (such as a spring) with displacement being a function of gravity. An increase or decrease in gravity will lengthen or shorten the main spring.

since gravimeters give us only relative gravity measurements, their accuracy is usually significantly higher than that of the absolute instruments.

A new design of gravimeters called
give around $5 \mu\text{Gal}$ accuracy, the best available.
But other available instruments provide some tenth
of μGal accuracy.

In practice, one tries to combine advantages of both techniques. This is done through the establishment of networks of gravity stations using both absolute and relative measurements. The stations with absolute gravity determination provide the ties between the points. The establishment of these global gravity networks has been coordinated internationally since the beginning of this century. The International Gravity Standardization Net 1971 (IGSN 71) is one of them.

When the absolute and relative observations are made and assessed for accuracy, an adjustment can be carried out. The adjustment results in the estimated values of gravity for all the stations, together with their accuracy estimate. In the IGSN 71, the accuracy of gravity at any station is on average 36 mGal . Once the global network is developed, it can be densified by acquiring new, usually relative, measurements. This is normally done by individual countries through the development of national gravity networks

(124)

adjusted to fit within the global network.

Needless to say, the determination of coordinates of the individual points is a part of the data acquisition process. Of these coordinates, the height is the most crucial.

After the detailed point gravity values have been adjusted, the surface gravity value g must be reduced to the geoid (g_0) using the appropriate gravity gradient

$$g_0 = g + \frac{\partial g}{\partial h} h \quad (256)$$

one can reduce the gravity anomaly too.

The above formula shows that surface gravity is corrected for the effect of the height h of the observation station. Clearly, every error made in the determination of the station height is transformed by the vertical gradient of gravity into an error in gravity anomaly Δg :

$$\sigma_{\Delta g}^2 = \sigma_g^2 + \left(\frac{\partial g}{\partial h} \right)^2 \sigma_h^2 \quad (257)$$

Finally, the computed point anomalies are then stored. In this terms they are used for various

geophysical and geodetic investigations. For geodetic application, however, a global coverage that is as homogeneous as possible is needed. For this reason, global anomaly files are used where mean anomalies are assigned to surface elements (cells) of a selected size delimited by geodetic coordinates φ, λ . These cells normally have following sizes: $5' \times 5'$, $20' \times 20'$, $30' \times 30'$, $1^\circ \times 1^\circ$, $5^\circ \times 5^\circ$ and $10^\circ \times 10^\circ$.

The mean anomalies $\overline{\Delta g}$ are evaluated from point anomalies Δg within each cell using different techniques. Here, the integral formula is given as an example:

$$\overline{\Delta g} = \frac{1}{A} \iint_S \Delta g \, ds \quad (258)$$

where A is the area of the cell S .

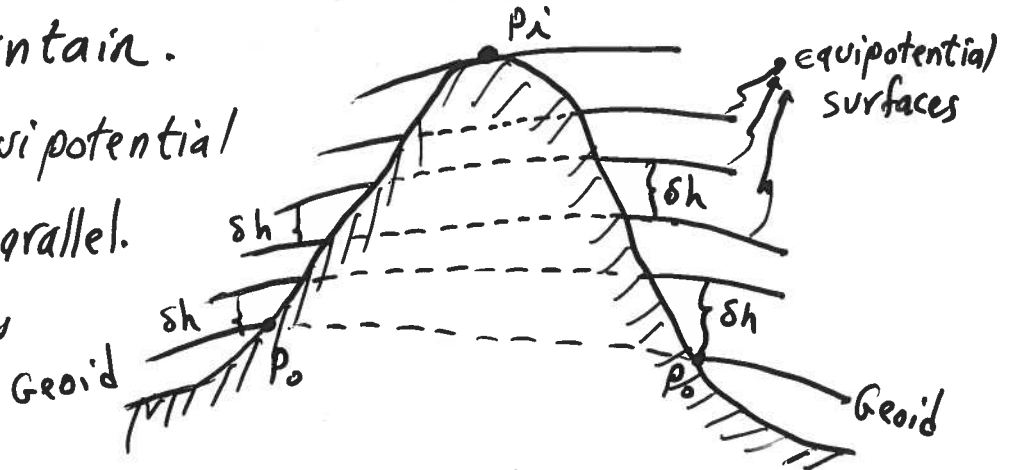
The main problem with global gravity files is the inhomogeneity of the data coverage. Depending on the size of the cell, a large percentage of the surface elements do not contain any gravity observations at all.

Height systems

levelling is also affected by the Earth's gravity field. To show this, let us consider a levelling line from point P_0 at sea-level to a point P_i on top of a mountain.

We know that equipotential surfaces are non-parallel.

Let us assume now that the first



levelling route goes up the left side and the second up the right side of the mountain. For each level setup, a height increment δh is obtained as the difference between backward and forward sightings on a level rod. The height of P_i

above sea level is then obtained by summing up the δh increments, first on the left side and then on the right side of the mountain. Clearly, two different values are obtained: there are greater separations δh between equipotential surfaces on the right side than on the left side. Which of the two values should be used for the height of P_i ?

This ambiguity can be eliminated only by converting the path-dependent results of levelling into unique path-independent height differences.

This can be solved using the Earth's gravity potential.

It is obvious that the difference between the potentials on two close together equipotential surfaces can be written as:

$$\delta W = -g\delta h \quad (259)$$

As only one equipotential surface passes through any point and, thus, there is only one value of potential W associated with each point.

Therefore, the gravity potential represents one way of defining a unique vertical position.

From now on we shall denote the observed (levelled) height differences by δl . Then if the δl is measured and the value of gravity g at the same location is known, the potential difference δW can then be computed from Eq. (259).

Instead of potential W_i of a point, we usually use the geopotential number C_i .

The geopotential number C_i is defined as the negative potential difference between point P_i and the geoid:

$$C_i = -(W_i - W_0) = \int_{P_0}^{P_i} g d\ell = \int_{P_0}^{P_i} g' dh' \quad (260)$$

where the integration is specified to proceed either along the terrain ($d\ell$) between the geoid and the point P_i or along the plumb line (dh') of P_i

similarly, the geopotential number difference

ΔC_{ij} between two points P_i and P_j

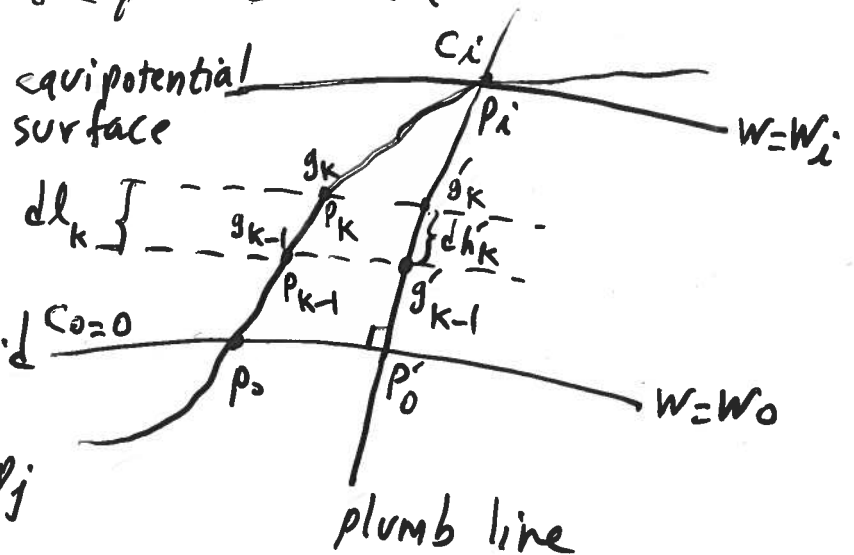
can be written as:

$$\Delta C_{ij} = \int_{P_i}^{P_j} g d\ell$$

(261)

The units for geopotential number adopted to be in kilogal meters. In this case the numerical value of the geopotential number in these units will be approximately equal to the height H of the point above sea level in meters. more precisely $C = 0.98 H$.

(129)



Geopotential numbers have several useful properties. They are unique for each point. Also, because the gravity potential is irrotational, the integral over a closed loop S is zero

$$\oint_S dC = \oint_C g dl = 0 \quad (262)$$

This evidently, is not true for levelled height differences $\oint_S dl \neq 0$.

Further, the geopotential numbers are positive above the geoid, zero at the geoid, negative below it, and constant on the same equipotential surface. Finally, a geopotential number difference can be evaluated from observations made only on the Earth's surface.

In particular, neither l nor g is known as a continuous function of position, then

$$\Delta C_{ij} = \sum_{k=i}^j \bar{g}_k \delta l_k \quad (263)$$

where

$$\bar{g}_k = \frac{1}{2} (g_{k-1} + g_k) \quad (264)$$

$\Delta\ell_k$ is the observed level difference between adjacent benchmarks, and g_k is the observed gravity value at the k th benchmark.

To eliminate the flaw of geopotential numbers not being expressed in length units, Dynamic Heights h^D have been introduced. They are obtained by dividing geopotential numbers by a constant reference gravity g_R :

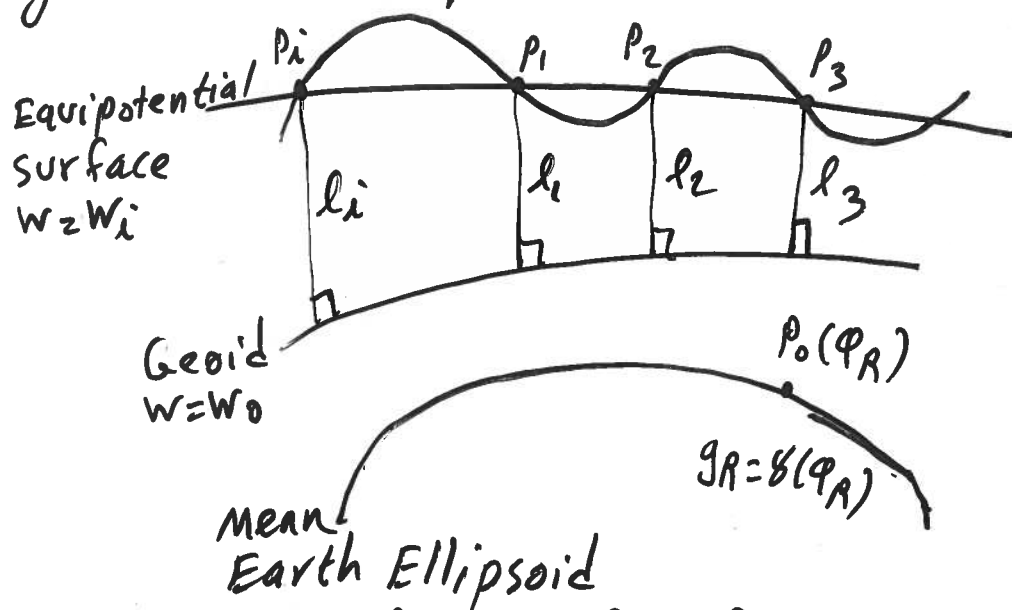
$$h_i^D = \frac{C_i}{g_R}$$

(265)

Here, g_R can be thought of as the value of normal gravity on the mean Earth Ellipsoid for a reference latitude ϕ_R selected so that g_R represents approximately the average gravity in the region of interest. The reference gravity value can be viewed as the scale factor needed to convert the geopotential number in units of potential to units of length. The datum for dynamic heights is still the geoid. Nevertheless one must be careful not to interpret the dynamic

(131)

height of a point as the geometrical distance between the geoid and the point



$$l_i \neq l_1 \neq l_2 \neq l_3 \quad \text{but} \quad H_i^D = H_1^D = H_2^D = H_3^D$$

since the dynamic heights of points on one equipotential surface are equal, the dynamic heights are as useful as the geopotential numbers for various (such as hydrological) projects requiring some knowledge of the physical Environment.

The dynamic height difference ΔH_{ij}^D between two points P_i, P_j is defined as:

$$\Delta H_{ij}^D = H_j^D - H_i^D = \frac{C_j}{g_R} - \frac{C_i}{g_R} = \frac{\Delta C_{ij}}{g_R} \quad (266)$$

An alternative formula for the dynamic height difference is obtained by expressing it as

$$\Delta h_{ij}^D = \Delta l_{ij} + DC_{ij} \quad (267)$$

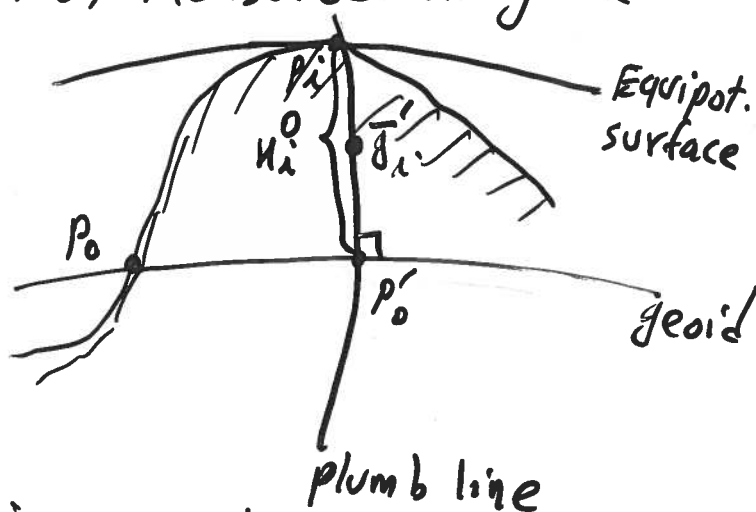
where Δl_{ij} is the measured levelled height difference and DC_{ij} is the dynamic correction which can be given by the following formula:

$$DC_{ij}^D = \sum_{k=i}^j \frac{\bar{g}_k - g_R}{g_R} \delta l_k \quad (268)$$

Many people need the geometrical concept of heights. The orthometric height H_i^0 of a point P_i is defined as the geometrical distance between the geoid and the point, measured along the plumb line of P_i .

The formula for H_i^0 can be written as:

$$H_i^0 = \int_{P_0}^{P_i} dh' \quad (269)$$



where the integration is carried out along the plumb line. substituting for dh' from Eq. (259):

$$H_i^0 = - \int_{P_0}^{P_i} \frac{dw}{g_i'} = \int_{P_0}^{P_i} \frac{dc}{g_i'} = \int_{P_0}^{P_i} \frac{g}{g_i'} dl \quad (270)$$

using mean value theorem for integrals

$$H_i^0 = \frac{1}{\bar{g}_i} \int_{P_0}^{P_i} g dl \quad (271)$$

where $\bar{g}_i$ is a value of gravity between the geoid and P_i . The $\bar{g}_i$ is the mean gravity along the plumb line of P_i in the integral sense. Then

$$H_i^0 = \frac{C_i^0}{\bar{g}_i}$$

(272)

It is practically impossible to determine $\bar{g}_i$ because g_i along the plumb line is not known since the density distribution within the Earth is not known. There are different approximate approaches with different assumptions made about the behaviour of the density within the Earth. The Helmert orthometric height is often used in practice. It is defined as:

$$H_i^H = \frac{C_i^H}{\bar{g}_i^H}$$

(273)

where the mean value $\bar{g}_i^H$ of gravity is

$$\bar{g}_i^H = g_i + 0.0424 H_i$$

(274)

where g_i is the gravity at P_i on the Earth's surface. The numerical coefficient follows directly from the use of Poincaré-Prey gravity gradient: $\bar{g}_i$ in this method is evaluated for the midpoint of the plumb line of P_i (134)

The major problem with orthometric height is that they are approximate depending on the kind of $\bar{g}_i$ used in the particular definition. Therefore it might be said that the orthometric heights are never really referred to the geoid but to another reference surface close to the geoid.

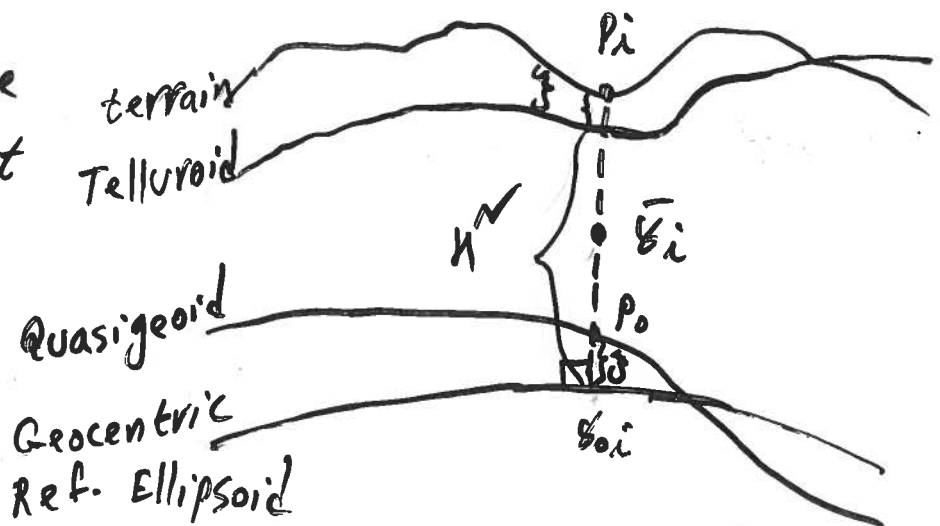
Also, points on the same geopotential surface (except on the geoid) do not have (generally) the same orthometric heights.

Normal heights H^N are suggested to replace the approximate orthometric heights. These are defined:

$$H_i^N = \frac{C_i}{\bar{g}_i}$$

(275)

where $\bar{g}_i$ is the normal counterpart of $\bar{g}_i$, i.e. the mean normal gravity along the plumb line of



P_i . The difference is that while the mean actual gravity $\bar{g}_i$ was computed from the downward continuation of the actual surface gravity g_i ,

(135)

the mean normal gravity $\bar{g}_i$ is computed from the upward continuation of normal gravity g_{oi} on the geocentric reference ellipsoid. The mean normal gravity is sought on the normal plumbline between points of height 0 (Ellipsoid) and height H^N (Telluroid).

To get $\bar{g}_i$ from g_{oi} , Molodenskiy uses accurate formula

$$\frac{\partial g}{\partial H} = -\frac{2g_0}{a} (1 + m + 2f \cos^2 \varphi)$$

for the vertical gradient of normal gravity resulting to Molodenskiy normal height. He has shown that his normal height can be regarded as a special kind of orthometric heights referred to a reference surface called quasigeoid.

Vignani proposed the derivation of $\bar{g}_i$ from g_{oi} be implemented using approximate vertical gradient of normal gravity $\frac{\partial g}{\partial H} = -0.3086 \frac{\text{mGal}}{\text{m}}$.

Then the Vignani normal height is given by:

$$H_i^V = \frac{C_i}{g_{oi} - 0.1543 H_i}$$

(276)

The same as dynamic heights an alternative

(136)

formula for orthometric as well as normal height can be written as:

$$\Delta H_{ij}^O = \Delta l_{ij} + DC_{ij}^O$$

$$\Delta H_{ij}^N = \Delta l_{ij} + DC_{ij}^N$$

(277)

where

$$DC_{ij}^O = \sum_{k=i}^j \frac{\bar{g}_k - g_R}{g_R} \delta l_k + \frac{\bar{g}_i - g_R}{g_R} H_i - \frac{\bar{g}_j - g_R}{g_R} H_j$$

$$DC_{ij}^N = \sum_{k=i}^j \frac{\bar{g}_k - g_R}{g_R} \delta l_k + \frac{\bar{g}_i - g_R}{g_R} H_i - \frac{\bar{g}_j - g_R}{g_R} H_j$$

The problem common to all three height (278)

systems used to be, and in some countries still is, the lack of available actual gravity data on the Earth's surface necessary for the evaluation of different heights. This problem was by-passed by using normal gravity $\gamma(\varphi, H)$ instead of actual g . Height differences based on normal gravity determined in this way do not depart excessively from the proper values when only adjacent benchmarks are considered. The effect of this approximation in networks of points is, however, more serious.

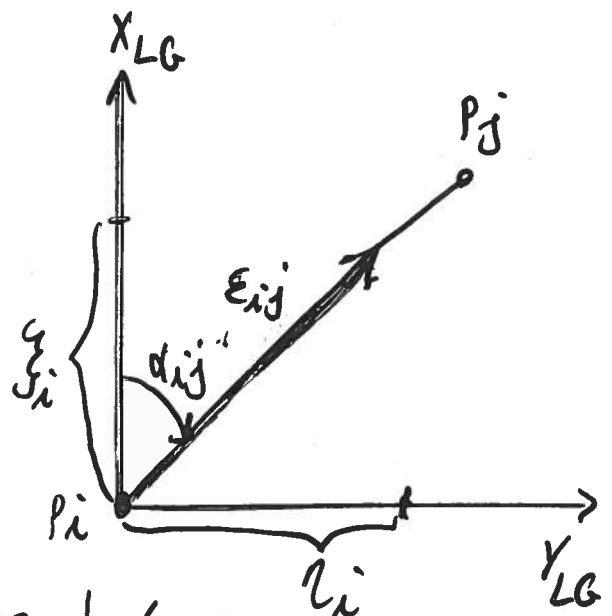
(137)

Non-gravimetric methods of geoid height determination

1) Astro-geodetic levelling

Geoidal height difference may be obtained from deflections of vertical by

$$\epsilon = - \frac{dN}{ds}$$

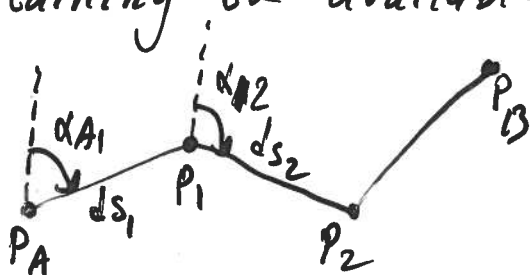


The geoidal height difference between two end points A, B of a curve on the geoid (which have the known deflection components) can be written as:

$$N_B - N_A = \int_A^B dN = - \int_A^B \epsilon ds = - \int_A^B (\xi \cos \alpha + \eta \sin \alpha) ds$$

In practice, the deflection components are known only at some geodetic control points, and the best which can be done is that to join the two end points by a traverse containing the available deflection points. (279)

The geoidal height difference is then evaluated from the approximate formula (138)



$$N_B - N_A = - \sum_{i=1}^n \epsilon_i ds_i$$

(280)

where

$$\epsilon_i = \frac{1}{2} [(\zeta_{i-1} + \zeta_i) \cos \alpha_{i-1,i} + (\eta_{i-1} + \eta_i) \sin \alpha_{i-1,i}]$$

This technique is also called Helmert levelling. To convert the geoidal height difference to geoidal heights, the geoidal height of at least one point must be known.

2) Gps-levelling geoidal height

Thanks to Gps, using a very simple formula

$$N = h - H$$

↓

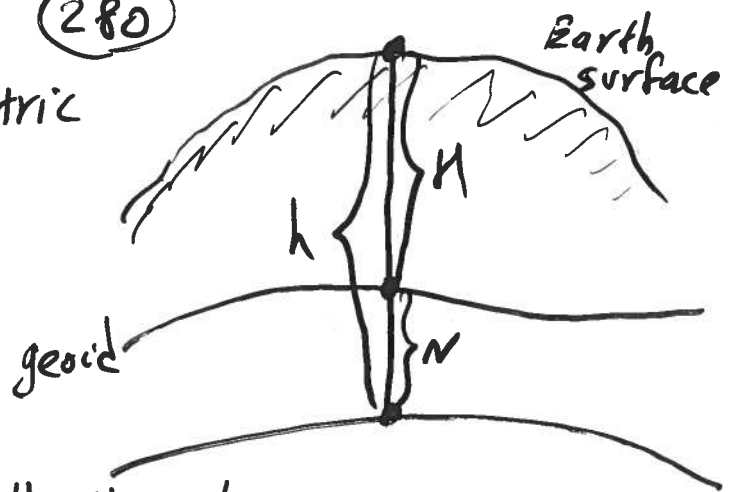
Gps ← Ellipsoidal height

orthometric height

(280)

The geoidal height N can be computed

easily, quickly, economically ellipsoid with less computational efforts.



This method is usually performed on the levelling lines where orthometric heights are measured and computed.

$$\Delta N = \Delta h - \Delta H$$

gives more accurate result

(281)

(139)

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